



Crown Digital: Brewing barista-quality, AI-driven coffee at scale

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GOOGLE CLOUD RESULTS

- ✓ Achieved 20-30% time savings on manual configurations, allowing the team to focus on innovation
- ✓ Accelerated deployment speeds by 20-30%, enabling faster rollouts of new solutions
- ✓ Empowered a lean tech team to manage global operations, supporting growth and scalability
- ✓ Built a strong foundation for AI-driven innovation, positioning Crown Digital as a pioneer in smart retail and robotics

Crown Digital's AI robot Ella uses Google Cloud's AI and machine learning solutions to deliver barista-quality coffee and support global expansion.

The new way to achieve global expansion with scalable AI



[Crown Digital](#) is a Singapore-based IoT startup revolutionizing the coffee experience with its AI-driven robot barista, Ella. Its initial infrastructure was constrained by manual configuration demands, slower deployment speeds, and insufficient resources for managing high-frequency data transfers across multiple locations. The lack of flexibility and scalability needed to process Ella's real-time data hampered the company's growth plans.

By leveraging Google [Firebase](#), Crown Digital created a unified, serverless backend for managing kiosk operations to maintain low latency across regions. They used [Google Cloud Storage](#) to handle Ella's vast data needs and maintain smooth, reliable access to operational and customer data across all locations. The new infrastructure minimized manual configurations, reduced deployment time, and gave the company the confidence to scale globally.

"We needed a platform that could support rapid scaling and global deployments with minimal manual effort. Google Cloud's integrated environment made it possible for us to run more with less, allowing our lean team to operate efficiently across borders," says Norman Ng, the Head of Technology at Crown Digital.

The company adopted [Vertex AI](#) and [Vision API](#) to train Ella in visual recognition tasks, such as detecting cup cleanliness, lid presence, and fill levels, which are vital for a premium, contactless coffee experience. With its innovative use of robotics, computer vision and machine learning, Crown Digital has become a pioneer in smart retail and robotics.

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Norman Ng

Head of Technology, Crown Digital



Why Google Cloud?

Crown Digital chose Google Cloud for its ease of deployment, integrated platform, and scalability, all essential for supporting rapid growth with minimal manual effort. Google Cloud’s suite of tools — including Firebase for mobile interactions, Vision API for visual recognition, and Vertex AI for machine learning — offered the specific functionalities needed to deliver efficient, data-driven customer experiences at scale.

"With Google Cloud, we can manage all our machines remotely, save time on deployments, and achieve our scalability goal. With this integrated environment, we can focus on growth and innovation rather than maintenance," adds CEO Keith Tan.

Google Cloud’s global reach and serverless options also influenced the decision. With Firebase, Crown Digital delivered low-latency services across multiple locations, unlocking the potential for scalable, contactless coffee service and setting the stage for future expansion.

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Keith Tan

CEO, Crown Digital

Crown Digital is a Singapore-based IoT startup revolutionizing the coffee experience with its AI-driven robot barista, Ella. Crown Digital leverages robotics, computer vision and machine learning to deliver barista-quality coffee through autonomous kiosks in high-traffic locations like airports, hospitals, and corporate offices. Crown Digital aims to make high-quality coffee accessible anytime, anywhere, while setting a new standard for innovation in the service industry.

Industry: Retail and Tech

Location: Singapore

Products: [Google Cloud](#), [Firebase](#), [Google Cloud Storage](#), [Vertex AI](#), [Vision API](#)

Why Google	Products and	Solutions	Resources	Engage
Choosing Google Cloud	pricing	Infrastructure modernization	Google Cloud Affiliate Program	Contact sales
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Modern Infrastructure Cloud	Google Workspace pricing	Application modernization	Google Cloud quickstarts	Become a Partner
Multicloud	See all products	Smart analytics	Google Cloud Marketplace	Events
Global infrastructure		Artificial Intelligence		Podcasts
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